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Potential Field-Based Sliding Surface Design and Its Application in Spacecraft Constrained Reorientation

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I. Introduction

DURING attitude maneuvers, there are usually specific constraints on the pointing of payloads onboard the spacecraft, including attitude forbidden zones and attitude mandatory zones. These attitude-constrained zones are usually defined as cones around specific unit vector directions in space. One typical example of attitude forbidden zones is that an optical-sensitive instrument (e.g., a telescope) is not allowed to point toward bright objects in space in order to avoid damage. One example of attitude mandatory zones is that the antenna of a spacecraft is required to keep pointing to the direction of the ground station for communication purposes during reorientation maneuvers.

Two main frameworks have been proposed to address the challenging problem of spacecraft constrained reorientation. Framework I is attitude planning and tracking, within which attitude constraints are considered by attitude planning and attitude controllers are implemented to track the designed rotational trajectory. Different attitude planning methods have been developed, such as the constraint monitor algorithm [1], the geometric method [2,3], the randomized exploring method [4], the discretization method [5,6], the recursive planning method based on rotational-path decomposition [7], the attitude planning method using gradient-based optimization [8], and the search-based planning in the rotation space SO(3) [9]. Though widely studied, attitude planning methods can be computationally expensive for the real-time onboard implementation. Framework II is to consider attitude constraints in the controller design, in which the most explored method is to combine artificial potential field (APF) with existing control methods. These APF-based methods include APF-based proportional and derivative (PD) controllers [10–12], APF-based backstepping controllers [13,14], APF-based sliding mode controllers [15], and APF-based kinematic steering laws [16]. APF-based methods are suitable for real-time onboard implementation because they are analytical. The inherent drawback of APF-based methods is the existence of local minima at which attractive and repulsive torques are balanced. The attitude of the spacecraft may be trapped at local minima rather than converging to the global minimum (i.e., the desired attitude). Only a limited number of works (e.g., [11,16]) have considered the problem of local minima.

The combination of sliding mode control (SMC) with APF was first proposed by Uktin et al. [17] and Guldner and Utkin [18] for collision avoidance of robots. The basic idea is to track the gradient of an APF using the SMC. Once the sliding mode is reached, the state of a robot will converge to the desired one and satisfy collision avoidance at the same time. The strategy of combining SMC with APF was explored by Shen et al. [15] for spacecraft reorientation with pointing constraints. However, compared with the strategy proposed by Uktin et al. [17] and Guldner and Utkin [18], the way of combining SMC with APF in [15] is different. To be specific, the sliding mode in [15] is the classical one for attitude maneuvers, in which the desired attitude can be achieved, whereas the attitude constraints cannot be guaranteed. Thus, the standard SMC method needs to be modified to consider attitude constraints. In [15], a classical Lyapunov function augmented with an attractive and a repulsive potential field was used to design the sliding mode controller.

Based on these observations, a question arises: How can we consider attitude constraints in the sliding surface design for attitude maneuvers by extending the idea of Utkin et al. [17] and Guldner and Utkin [18]? The main purpose of this Note is to answer this question. The main contributions are as follows:

1) By introducing APF into the design process of the sliding surface, this Note presents new results of sliding surfaces using different attitude parameterizations for spacecraft attitude maneuvers. The existing classical results of sliding surfaces in the literature are shown to be special cases of the obtained new results.

2) By applying the proposed sliding surface design principle, this Note designs a new sliding surface for spacecraft reorientation with attitude constraints. The novelty of the proposed sliding surface is that, once the sliding mode is reached, the attitude of the spacecraft will converge to the desired one while satisfying attitude constraints automatically even under external disturbances. Moreover, the existence and the characteristics of critical points of the potential field in the sliding mode are analyzed. As attitude constraints are considered in the sliding surface design, a standard controller design method of SMC can be implemented for the constrained reorientation without any adjustment.

This Note is organized as follows. In Sec. II, dynamic equations of spacecraft rotational motion and attitude constraints on spacecraft reorientation maneuvers are presented. In Sec. III, based on a potential field-based sliding surface design principle, new results of sliding surfaces for spacecraft attitude maneuvers using different attitude parameterizations are presented. In Sec. IV, the proposed design principle is applied to the sliding surface design for spacecraft constrained reorientation and a sliding mode controller is developed based on the proposed sliding surface. The stability of the closed-loop system under the proposed controller is proven using the Lyapunov method. The existence and the characteristics of critical points of the potential field in the sliding mode are analyzed. Section V presents simulation results to validate the proposed controller for the constrained reorientation. Conclusions are drawn in Sec. VI.

II. Preliminaries

This section presents kinematic and dynamic equations of spacecraft rotational motion using different attitude parameterizations. Attitude constraints are described and mathematically formulated using unit quaternions.

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A. Rotational Kinematics Based on Different Parameterizations and Dynamics

The kinematics using unit quaternions is given as [19]

$$\dot{\boldsymbol{q}} = \frac{1}{2}\boldsymbol{q} \otimes \boldsymbol{\omega} \tag{1}$$

where $\boldsymbol{q} = (q_0, \bar{q})$ is a unit quaternion representing the attitude, with $q_0 \in \mathbb{R}$ being its scalar part and $\bar{q} = [q_1, q_2, q_3]^T \in \mathbb{R}^3$ being its vector part; $\boldsymbol{\omega} = (0, \bar{\omega})$ is the quaternion form of the angular velocity $\bar{\omega} = [\omega_1, \omega_2, \omega_3]^T \in \mathbb{R}^3$; and $\otimes$ is quaternion multiplication. Given two quaternions, $\boldsymbol{a} = (a_0, \bar{a})$ and $\boldsymbol{b} = (b_0, \bar{b})$, quaternion multiplication is calculated as $\boldsymbol{a} \otimes \boldsymbol{b} = (a_0 b_0 - \bar{a} \cdot \bar{b}, a_0\bar{b} + b_0\bar{a} + \bar{a} \times \bar{b})$. The conjugate of quaternion is defined as $\boldsymbol{a}^* = (a_0, -\bar{a})$. The vector part of quaternion is defined as $\text{vec}(\boldsymbol{a}) = \bar{a}$.

The kinematics using Rodrigues parameters $\bar{\rho} \in \mathbb{R}^3$ is as follows [19]:

$$\dot{\bar{\rho}} = T(\bar{\rho})\bar{\omega} \tag{2}$$

where $\bar{\rho} = [\rho_1, \rho_2, \rho_3]^T$ is related to unit quaternion by $\bar{\rho} = \bar{q}/q_0$ [19] and the matrix $T(\bar{\rho})$ is defined as $T(\bar{\rho}) \triangleq (1/2)[I_{3\times3} + \bar{\rho}\bar{\rho}^T + \bar{\rho}^\times]$ with $I_{3\times3} \in \mathbb{R}^{3\times3}$ being the identity matrix. $\bar{\rho}^\times$ is the cross-product matrix, which is defined as

$$\bar{\rho}^\times = \begin{bmatrix} 0 & -\rho_3 & \rho_2 \\ \rho_3 & 0 & -\rho_1 \\ -\rho_2 & \rho_1 & 0 \end{bmatrix} \tag{3}$$

The kinematics using modified Rodrigues parameters $\bar{\sigma} \in \mathbb{R}^3$ is summarized as [19]

$$\dot{\bar{\sigma}} = F(\bar{\sigma})\bar{\omega} \tag{4}$$

where $\bar{\sigma} \in \mathbb{R}^3$ is related to unit quaternion by $\bar{\sigma} = \bar{q}/(1 + q_0)$ [19] and the matrix $F(\bar{\sigma})$ is defined as $F(\bar{\sigma}) \triangleq (1/4)[(1 - \bar{\sigma}^T\bar{\sigma})I_{3\times3} + 2\bar{\sigma}^\times + 2\bar{\sigma}\bar{\sigma}^T]$ with $\bar{\sigma}^\times$ being the cross-product matrix.

The dynamic equation is given as [19]

$$I\dot{\bar{\omega}} = -\bar{\omega}^\times I\bar{\omega} + \bar{u} + \bar{d} \tag{5}$$

where $I \in \mathbb{R}^{3\times3}$ is the inertia matrix of the spacecraft, $\bar{u} \in \mathbb{R}^3$ is the control torque, $\bar{d} \in \mathbb{R}^3$ is the external disturbance torque, and $\bar{\omega}^\times$ is the cross-product matrix.

B. Attitude Constraints Based on Unit Quaternions

Both attitude forbidden zones and attitude mandatory zones are considered in this Note. These attitude-constrained zones are usually defined as cones around specific unit vector directions in space. An attitude forbidden zone is a cone that the boresight vector of certain instrument (e.g., a telescope) of a spacecraft is required to avoid during its reorientation. An attitude mandatory zone is a cone, in which the boresight vector of certain instrument (e.g., an antenna) of a spacecraft is required to stay.

Figure 1 illustrates a sketch of a spacecraft equipped with a telescope and an antenna, and its related attitude-constrained zones. The red- and blue-shaded cones represent the attitude forbidden zone (indicated by the subscript F) related to the telescope and the attitude mandatory zone (indicated by the subscript M) related to the antenna, respectively; $\bar{r}_F^B$ and $\bar{r}_M^B$, both expressed in the body frame as indicated by the superscript B, denote unit boresight vectors of the telescope and the antenna, respectively; $\bar{n}_F^I$ and $\bar{n}_M^I$, both expressed in the inertial frame as indicated by the superscript I, represent the unit direction vector of the bright object to avoid and the unit direction vector of the desired direction for the communication, respectively; θ_F and θ_M are half angles of cones representing the attitude forbidden zone and the attitude mandatory zone, respectively.

The unit boresight vector of the telescope, $\bar{r}_F^B$, is required to stay outside the red-shaded attitude forbidden zone characterized by $\bar{n}_F^I$ and θ_F. This requirement can be mathematically formulated as

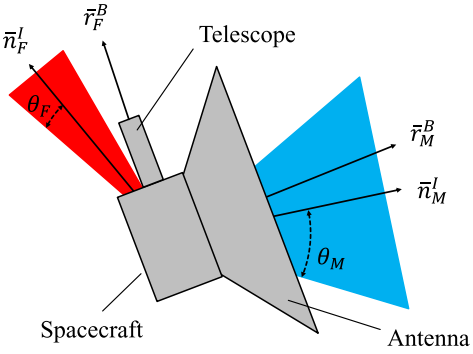


Fig. 1 Spacecraft sketch with attitude-constrained zones.

$$\bar{r}_F^I \cdot \bar{n}_F^I - \cos\theta_F < 0 \tag{6}$$

where $\bar{r}_F^I$ is the expression of $\bar{r}_F^B$ in the inertial frame, which can be calculated based on the transformation [19] $\boldsymbol{r}_F^I = \boldsymbol{q} \otimes \boldsymbol{r}_F^B \otimes \boldsymbol{q}^*$, with $\boldsymbol{r}_F^B = (0, \bar{r}_F^B)$ and $\boldsymbol{r}_F^I = (0, \bar{r}_F^I)$ denoting quaternion forms of $\bar{r}_F^B$ and $\bar{r}_F^I$, respectively.

Based on unit quaternions, the requirement of the attitude forbidden zone in Eq. (6) can be rewritten as [13]

$$\boldsymbol{q}^T M_F \boldsymbol{q} < 0 \tag{7}$$

with

$$M_F = \begin{bmatrix} m_1 & \bar{m}_2^T \\ \bar{m}_2 & A \end{bmatrix} \tag{8}$$

where $m_1 = \bar{r}_F^B \cdot \bar{n}_F^I - \cos\theta_F$, $\bar{m}_2 = \bar{r}_F^B \times \bar{n}_F^I$, and $A = \bar{r}_F^B(\bar{n}_F^I)^T + \bar{n}_F^I(\bar{r}_F^B)^T - (\bar{r}_F^B \cdot \bar{n}_F^I + \cos\theta_F)I_{3\times3}$. Note that M_F is a symmetric matrix, i.e., $M_F = M_F^T$.

To keep the continuous communication with the ground station on Earth, the unit boresight vector of the antenna, $\bar{r}_M^B$, is required to stay in the blue-shaded attitude mandatory zone characterized by $\bar{n}_M^I$ and θ_M. Similar to the formulation of the attitude forbidden zone, the requirement of the attitude mandatory zone can be written as [13]

$$\boldsymbol{q}^T M_M \boldsymbol{q} > 0 \tag{9}$$

where M_M has the same form as M_F in Eq. (8), with $\bar{r}_F^B$, $\bar{n}_F^I$, and θ_F being replaced by $\bar{r}_M^B$, $\bar{n}_M^I$, and θ_M, respectively.

It should be mentioned that, because attitude constraints considered in this Note are assumed to be fixed in the inertial frame, both M_F and M_M are constant matrices once parameters of attitude constraints are set. In addition, it has been shown by Lee and Mesbahi [13] that $-1 < (1/2)\boldsymbol{q}^T M_F \boldsymbol{q} < 0$ and $0 < (1/2)\boldsymbol{q}^T M_M \boldsymbol{q} < 1$ hold for attitude forbidden zones and attitude mandatory zones, respectively.

III. Potential Field-Based Sliding Surface Design for Spacecraft Attitude Maneuvers

This section explains the basic idea of the potential field-based sliding surface design and presents new results of sliding surfaces using different attitude parameterizations based on the proposed design methodology.

A. Methodology

The potential field-based sliding surface design methodology in this Note is inspired by sliding surface design in [17,18] and the sliding surface design using transformation to reduced form in [20]. In the potential field-based sliding surface design methodology, a potential function V_p is first introduced. V_p is designed so that it only depends on the spacecraft attitude with the property that $V_p \geq 0$ for all attitudes and $V_p = 0$ holds if and only if at the desired attitude.

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Note that the designed potential function can also be regarded as a Lyapunov function for the sliding mode phase. Then, the sliding surface design is approached in a backward manner. To be specific, the expression of $\dot{V}_p$ is investigated and some constrained motions are chosen to ensure $\dot{V}_p$ to be negative definite. These constrained motions are actually candidates of the sliding surface.

B. Sliding Surface Design: Results Based on Different Attitude Parameterizations

The proposed design methodology is applied in spacecraft attitude tracking maneuvers. The application of the design principle to the sliding surface design based on unit quaternion is described in detail.

Let the APF V_p be a function of the error unit quaternion $\boldsymbol{q}_e = (q_{e0}, \bar{q}_e)$, where $\bar{q}_e = [q_{e1}, q_{e2}, q_{e3}]^T$ and $\boldsymbol{q}_e = \boldsymbol{q}_d^* \otimes \boldsymbol{q}$, with $\boldsymbol{q}_d$ denoting the desired attitude. V_p is designed to have the property that $V_p \geq 0$ and $V_p = 0$ if and only if $\boldsymbol{q}_e = \pm\mathbf{1}$, where $\mathbf{1} \triangleq (1, \bar{0})$ denotes the identity quaternion. (Note that $\mathbf{1}$ and $-\mathbf{1}$ denote the same attitude.) The design of the sliding surface is based on the following observation: Taking the time derivative of V_p and using the relative kinematic equation $\dot{\boldsymbol{q}}_e = (1/2)\boldsymbol{q}_e \otimes \boldsymbol{\omega}_e$ [21] with $\boldsymbol{\omega}_e = \boldsymbol{\omega} - \boldsymbol{\omega}_d$ (note that both $\boldsymbol{\omega}$ and $\boldsymbol{\omega}_d$ are expressed in the body frame) yields

$$\begin{aligned}\dot{V}_p &= (\nabla V_p)^T \dot{\boldsymbol{q}}_e \\ &= \frac{1}{2}(\nabla V_p)^T(\boldsymbol{q}_e \otimes \boldsymbol{\omega}_e) \\ &= \frac{1}{2}\boldsymbol{\omega}_e^T(\boldsymbol{q}_e^* \otimes \nabla V_p) \\ &= \frac{1}{2}\bar{\omega}_e^T \mathrm{vec}(\boldsymbol{q}_e^* \otimes \nabla V_p)\end{aligned} \tag{10}$$

where the property that $\boldsymbol{a}^T(\boldsymbol{b} \otimes \boldsymbol{c}) = \boldsymbol{c}^T(\boldsymbol{b}^* \otimes \boldsymbol{a})$ is used [13] and the gradient of V_p is defined as

$$\nabla V_p \triangleq \frac{\partial V_P}{\partial \boldsymbol{q}_e} = \left[\frac{\partial V_p}{\partial q_{e0}}, \frac{\partial V_p}{\partial q_{e1}}, \frac{\partial V_p}{\partial q_{e2}}, \frac{\partial V_p}{\partial q_{e3}}\right]^T$$

Let $\bar{\omega}_e = -K_\alpha \mathrm{vec}(\boldsymbol{q}_e^* \otimes \nabla V_p)$, with $K_\alpha \in \mathbb{R}^{3\times3}$ being a diagonal positive matrix, and then it yields $\dot{V}_p = -(1/2)\bar{\omega}_e^T K_\alpha^{-1} \bar{\omega}_e \leq 0$. When V_p decreases to zero, the desired attitude is reached. Based on these observations, the sliding surface is designed as

$$\bar{s} = \bar{\omega}_e + K_\alpha \mathrm{vec}(\boldsymbol{q}_e^* \otimes \nabla V_p) \tag{11}$$

where $K_\alpha \in \mathbb{R}^{3\times3}$ is a diagonal positive matrix.

The potential function can be chosen as the following quadratic function:

$$V_p = \frac{1}{2}\|\boldsymbol{q}_e - \delta(q_{e0})\mathbf{1}\|^2 \tag{12}$$

where $\|\cdot\|$ denotes the norm of quaternions; q_{e0} is the scalar element of $\boldsymbol{q}_e$; and the term $\delta(q_{e0})$ with the property that $\delta(q_{e0}) = 1$ if $q_{e0} \geq 0$ and $\delta(q_{e0}) = -1$ if $q_{e0} < 0$ is introduced to avoid the unwinding phenomenon. (When the unwinding phenomenon occurs, the spacecraft will not rotate in the shortest path toward the desired attitude [11,22].) By noting that $(1/2)\|\boldsymbol{q}_e - \delta(q_{e0})\mathbf{1}\|^2 = (1/2)[\boldsymbol{q}_e - \delta(q_{e0})\mathbf{1}]^T[\boldsymbol{q}_e - \delta(q_{e0})\mathbf{1}] = 1 - \delta(q_{e0})\boldsymbol{q}_e^T\mathbf{1}$, the gradient of potential function is $\nabla V_p = (\partial V_p/\partial \boldsymbol{q}_e) = -\delta(q_{e0})\mathbf{1}$ and $\nabla^2 V_p = 0_{3\times3}$. Thus, the sliding surface in Eq. (11) can be simplified to

$$\begin{aligned}\bar{s} &= \bar{\omega}_e + K_\alpha \mathrm{vec}(\boldsymbol{q}_e^* \otimes (-\delta(q_{e0})\mathbf{1})) \\ &= \bar{\omega}_e + \delta(q_{e0})K_\alpha \mathrm{vec}(\boldsymbol{q}_e) \\ &= \bar{\omega}_e + \delta(q_{e0})K_\alpha \bar{q}_e\end{aligned} \tag{13}$$

Note that the sliding surface in Eq. (13) is the same as the classical sliding surface for the attitude tracking in [23].

Similarly, the sliding mode design based on Rodrigues parameters and modified Rodrigues parameters can be obtained (see Appendices A and B). These main results are summarized in Table 1. Note that, in Table 1, for unit quaternion, $\bar{\omega}_e \triangleq \bar{\omega} - \bar{\omega}_d$, $\boldsymbol{q}_e \triangleq \boldsymbol{q}_d^* \otimes \boldsymbol{q}$, and $\nabla V_p \triangleq \partial V_p/\partial \boldsymbol{q}_e$; for Rodrigues parameters, $\bar{\omega}_e \triangleq \bar{\omega} - T^{-1}(\bar{\rho})T(\bar{\rho}_d)\bar{\omega}_d$, $\bar{\rho}_e \triangleq \bar{\rho} - \bar{\rho}_d$, and $\nabla V_p \triangleq \partial V_p/\partial \bar{\rho}_e$; for modified Rodrigues parameters, $\bar{\omega}_e \triangleq \bar{\omega} - F^{-1}(\bar{\sigma})F(\bar{\sigma}_d)\bar{\omega}_d$, $\bar{\sigma}_e \triangleq \bar{\sigma} - \bar{\sigma}_d$, and $\nabla V_p \triangleq \partial V_p/\partial \bar{\sigma}_e$. Note that the special forms in Table 1 are the same as the results in [23–26], respectively.

Sliding surfaces for attitude regulation based on the proposed design methodology can be obtained by modifying results in Table 1 as the attitude regulation is a special case of the attitude tracking. Note that, with different designs of the potential field, different sliding surfaces for attitude maneuvers can be obtained. Moreover, by exploring the similarity between quaternions and dual quaternions, the sliding surface based on dual quaternions for relative pose (i.e., relative position and relative attitude) control in [27] can be obtained using the proposed sliding surface design methodology.

IV. Application of Potential Field-Based Sliding Surface Design in Spacecraft Reorientation with Attitude Constraints

In this section, we present the application of the proposed sliding mode design principle in spacecraft reorientation with attitude constraints. Unit quaternion is used as the attitude parameterization.

A. Sliding Surface for Spacecraft Constrained Reorientation

As for the reorientation case, based on the sliding surface in Eq. (11), the sliding surface is designed as

$$\bar{s} = \bar{\omega} + \alpha \mathrm{vec}(\boldsymbol{q}^* \otimes \nabla V_p) \tag{14}$$

where $\alpha \in \mathbb{R}$ and $\alpha > 0$.

The attitude constraints are considered in the design of the APF. Inspired by the structure of the potential field by Hu et al. [11], the following potential field is designed:

$$V_p = \beta_1\|\boldsymbol{q} - \delta(q_{e0})\boldsymbol{q}_d\|^2 + \beta_2\|\boldsymbol{q} - \delta(q_{e0})\boldsymbol{q}_d\|^2(f_F + f_M) \tag{15}$$

Table 1 Summary of sliding surfaces

Attitude parameterization	General form of sliding surface	Special form of sliding surface with quadratic form of V_p
Unit quaternion	$\bar{s} = \bar{\omega}_e + K_\alpha \mathrm{vec}(\boldsymbol{q}_e^* \otimes \nabla V_p)$	$\bar{s} = \bar{\omega}_e + \delta(q_{e0})K_\alpha \bar{q}_e$ (with $V_p = (1/2)\|\boldsymbol{q}_e - \delta(q_{e0})\mathbf{1}\|^2$)
Rodrigues parameters	$\bar{s}_1 = \bar{\omega}_e + T^{-1}(\bar{\rho})K_\alpha \nabla V_p$	$\bar{s}_1 = \bar{\omega}_e + T^{-1}(\bar{\rho})K_\alpha \bar{\rho}_e$ (with $V_p = (1/2)\|\bar{\rho}_e\|^2 = (1/2)\bar{\rho}_e^T\bar{\rho}_e$)
	$\bar{s}_2 = \bar{\omega}_e + c\nabla V_p$	$\bar{s}_2 = \bar{\omega}_e + \Lambda'\bar{\rho}_e$ (with $V_p = (1/2)\bar{\rho}_e^T\Lambda\bar{\rho}_e$ and $\Lambda' = c\Lambda$)
Modified Rodrigues parameters	$\bar{s} = \bar{\omega}_e + F^{-1}(\bar{\sigma})K_\alpha \nabla V_p$	$\bar{s} = \bar{\omega}_e + F^{-1}(\bar{\sigma})K_\alpha \bar{\sigma}_e$ (with $V_p = (1/2)\|\bar{\sigma}_e\|^2 = (1/2)\bar{\sigma}_e^T\bar{\sigma}_e$)

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where $\boldsymbol{q}_d$ is the desired attitude, which is a constant unit quaternion for the reorientation problem; the term $\delta(q_{e0})$ has the same definition as in Eq. (12); $\beta_1 > 0$ and $\beta_2 > 0$ are parameters to be set; and barrier potential fields $f_F(\boldsymbol{q})$ and $f_M(\boldsymbol{q})$ are defined as [13]

$$f_F = \sum_{j=1}^{m}\sum_{i=1}^{n} -k_1 \ln\left(-\frac{\boldsymbol{q}^T M_{Fi}^j \boldsymbol{q}}{2}\right) \tag{16}$$

and

$$f_M = -k_2 \ln\left(\frac{\boldsymbol{q}^T M_M \boldsymbol{q}}{2}\right) \tag{17}$$

where $k_1 > 0$ and $k_2 > 0$.

For convenience, define $f = f_F + f_M$. By noting that $\|\boldsymbol{q} - \delta(q_{e0})\boldsymbol{q}_d\|^2 = 2 - 2\delta(q_{e0})\boldsymbol{q}_d^T\boldsymbol{q}$, ∇V_p can be calculated as

$$\nabla V_p = -2\delta(q_{e0})(\beta_1 + \beta_2 f)\boldsymbol{q}_d + 2\beta_2(1 - \delta(q_{e0})\boldsymbol{q}_d^T\boldsymbol{q})\frac{\partial f}{\partial \boldsymbol{q}} \tag{18}$$

where $(\partial f/\partial \boldsymbol{q}) = (\partial f_F/\partial \boldsymbol{q}) + (\partial f_M/\partial \boldsymbol{q})$. The calculation of $\partial f_F/\partial \boldsymbol{q}$ and $\partial f_M/\partial \boldsymbol{q}$ is detailed in Appendix C.

B. Analysis of Critical Points of Potential Field in Sliding Mode

When the state reaches and stays on the sliding surface, i.e., $\bar{s} = \bar{\omega} + \alpha \text{vec}(\boldsymbol{q}^* \otimes \nabla V_p) = \bar{0}$, one has $\dot{V}_p = -(1/2\alpha)\bar{\omega}^T\bar{\omega} \le 0$. For $V_p = 0$, the desired attitude is reached. However, it is possible that $\bar{\omega} = -\alpha \text{vec}(\boldsymbol{q}^* \otimes \nabla V_p) = \bar{0}$ when $\boldsymbol{q} \neq \delta(q_{e0})\boldsymbol{q}_d$, which leads to the problem of local minima [11].

Figure 2 shows the possible time histories of V_p with three different cases of critical points of the potential field in the sliding mode. It should be mentioned that, in this Note, critical points of the potential field mean the attitudes at which $\dot{V}_p = 0$ holds. When $\dot{V}_p = 0$ occurs at the desired attitude, this corresponds to the global minimum (as shown in Fig. 2a). When $\dot{V}_p = 0$ occurs at an attitude that is not the desired one, the attitude can be either a local minimum or a saddle point. When a critical point is a local minimum, the attitude of the spacecraft will be trapped at this local minimum (as shown in Fig. 2b). When a critical point is a saddle point, any small disturbances or measurement noises can drive the attitude away from this saddle point (as shown in Fig. 2c). Hu et al. [11] presented a detailed quantitative analysis of the problem of local minima. However, it is difficult to extend their method to the case with complex attitude constraints. In this Note, we present a qualitative analysis of critical points as follows.

For $\text{vec}(\boldsymbol{q}^* \otimes \nabla V_p) = \bar{0}$, there are two cases for the solution: one case is $\boldsymbol{q}^* \otimes \nabla V_p = \boldsymbol{0}$, with $\boldsymbol{0} \triangleq (0, \bar{0})$, and the other is $\boldsymbol{q}^* \otimes \nabla V_p = \lambda \boldsymbol{1}$, where $\lambda \in \mathbb{R}$ and $\lambda \neq 0$.

Case 1: $\boldsymbol{q}^* \otimes \nabla V_p = \boldsymbol{0}$

The following proposition is presented for the analysis.

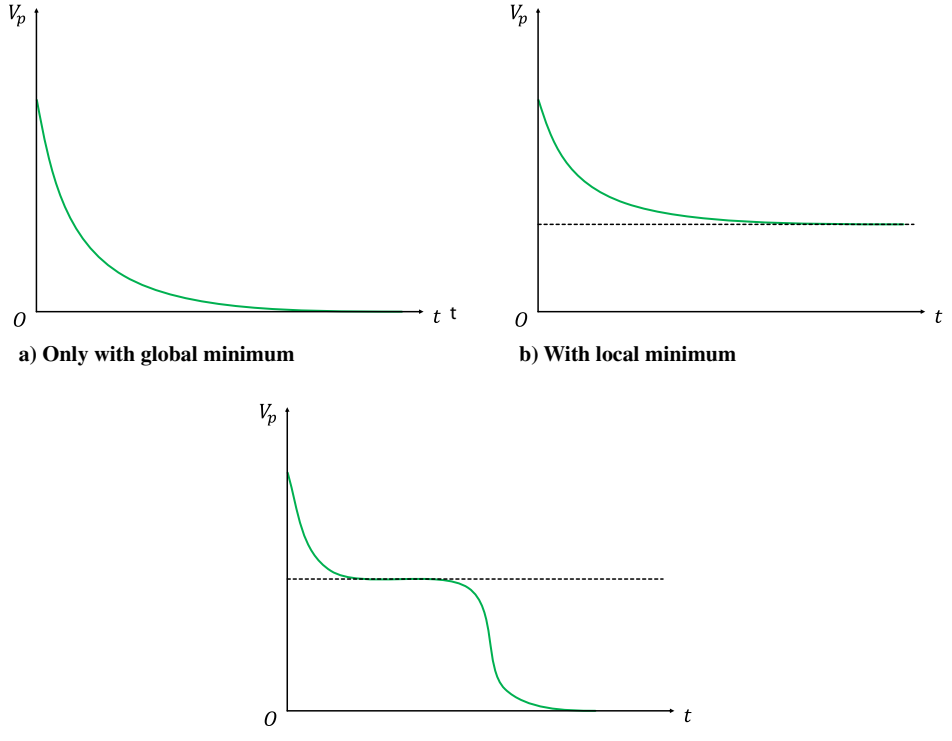
Proposition 1: $\boldsymbol{q}^* \otimes \nabla V_p = \boldsymbol{0}$ yields $\nabla V_p = \boldsymbol{0}$. Furthermore, based on Eq. (18), $\nabla V_p = \boldsymbol{0}$ yields $1 - \delta(q_{e0})\boldsymbol{q}_d^T\boldsymbol{q} \neq 0$, and hence $\boldsymbol{q} \neq \delta(q_{e0})\boldsymbol{q}_d$.

Proof: See Appendix D.

When $\boldsymbol{q}^* \otimes \nabla V_p = \boldsymbol{0}$ holds, based on Proposition 1, Eq. (18) can be rewritten as

$$\frac{\partial f}{\partial \boldsymbol{q}} = \frac{\delta(q_{e0})}{1 - \delta(q_{e0})\boldsymbol{q}_d^T\boldsymbol{q}}\left(\frac{\beta_1}{\beta_2} + f\right)\boldsymbol{q}_d \tag{19}$$

As indicated by Proposition 1, the attitude cannot reach the desired one. Thus, the solution (or solutions) to Eq. (19) can either be a local minimum or a saddle point of the potential field in the sliding mode. As shown in Eq. (19), the solution is related to the configuration of the potential field since f, $\partial f/\partial \boldsymbol{q}$, β_1, and β_2 are involved. But it is difficult to obtain an analytical solution to Eq. (19). Here, we present a qualitative analysis to show the effect of the ratio β_1/β_2 on the locations of critical points. Based on Eqs. (C1) and (C2), and $(\partial f/\partial \boldsymbol{q}) = (\partial f_F/\partial \boldsymbol{q}) + (\partial f_M/\partial \boldsymbol{q})$, when $\boldsymbol{q}$ is in the feasible set (i.e., the set of attitudes at which attitude constraints are satisfied), $\partial f/\partial \boldsymbol{q}$ is well defined. Consider an extreme case where we increase the ratio β_1/β_2 (e.g., increasing β_1 while fixing β_2) to ∞. According to Eq. (19), the value of $\partial f/\partial \boldsymbol{q}$ will increase to ∞, which, according to Eqs. (C1) and (C2), is actually the consequence when $\boldsymbol{q}^T M_{F_i}^j \boldsymbol{q}$ and/or $\boldsymbol{q}^T M_M \boldsymbol{q}$ approach 0. Thus, it means that the solution attitude $\boldsymbol{q}$ to the



a) Only with global minimum **b) With local minimum**

c) With saddle point

Fig. 2 Illustration of V_p with different critical points.

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equation $\nabla V_p = \mathbf{0}$ will approach boundaries of attitude-constrained zones when the ratio β_1/β_2 increases to a large value.

Case 2: $\boldsymbol{q}^* \otimes \nabla V_p = \lambda \mathbf{1}$ ($\lambda \in \mathbb{R}$ and $\lambda \neq 0$)

In this case, $\boldsymbol{q}^* \otimes \nabla V_p = \lambda \mathbf{1}$ yields $\nabla V_p = \lambda \boldsymbol{q}$. Furthermore, $\|\nabla V_p\| = |\lambda|$ holds as $\boldsymbol{q}$ is a unit quaternion. Based on Eq. (18), $\boldsymbol{q}^* \otimes \nabla V_p$ is calculated as

$$\boldsymbol{q}^* \otimes \nabla V_p = -2\delta(q_{e0})(\beta_1 + \beta_2 f)(\boldsymbol{q}^* \otimes \boldsymbol{q}_d) + 2\beta_2(1 - \delta(q_{e0})\boldsymbol{q}_d^T\boldsymbol{q})\left(\boldsymbol{q}^* \otimes \frac{\partial f}{\partial \boldsymbol{q}}\right) \tag{20}$$

It can be verified that $\boldsymbol{q} = \delta(q_{e0})\boldsymbol{q}_d$ leads to $\boldsymbol{q}^* \otimes \nabla V_p = -2\delta(q_{e0})(\beta_1 + \beta_2 f)\mathbf{1}$, which means that the desired attitude satisfies the condition that $\boldsymbol{q}^* \otimes \nabla V_p = \lambda \mathbf{1}$. Depending on the configuration of the attitude-constrained zones (namely, the terms f_F and f_M) and the weight parameters set (namely, parameters β_1 and β_2), it is possible that $\boldsymbol{q}^* \otimes \nabla V_p = \lambda \mathbf{1}$ holds for certain $\boldsymbol{q} \neq \delta(q_{e0})\boldsymbol{q}_d$. This also leads to local minima or saddle points of the potential field. By collecting terms, Eq. (20) can be rewritten as

$$\boldsymbol{q}^* \otimes \nabla V_p = -2\beta_2\left(\frac{\beta_1}{\beta_2} + f\right)\left[\delta(q_{e0})(\boldsymbol{q}^* \otimes \boldsymbol{q}_d) - \frac{1 - \delta(q_{e0})\boldsymbol{q}_d^T\boldsymbol{q}}{(\beta_1/\beta_2) + f}\left(\boldsymbol{q}^* \otimes \frac{\partial f}{\partial \boldsymbol{q}}\right)\right] \tag{21}$$

When the ratio β_1/β_2 increases to a very large value, compared with the first term in the square bracket in Eq. (21), the second term vanishes, which yields the following:

$$\boldsymbol{q}^* \otimes \nabla V_p \approx -2\beta_2\left(\frac{\beta_1}{\beta_2} + f\right)\delta(q_{e0})(\boldsymbol{q}^* \otimes \boldsymbol{q}_d) \tag{22}$$

Thus, $\delta(q_{e0})\boldsymbol{q}_d$ will be the unique solution to $\boldsymbol{q}^* \otimes \nabla V_p = \lambda \mathbf{1}$ when the ratio β_1/β_2 increases to a very large value.

From the previous analysis, it can be concluded that locations of local minima or saddle points of the potential field can be adjusted by changing the ratio β_1/β_2, which benefits from the introduction of the term $\beta_1\|\boldsymbol{q} - \delta(q_{e0})\boldsymbol{q}_d\|^2$ in the potential field. The analysis presented in this subsection can be used to guide the setting of parameters of the potential field.

C. Sliding Mode Controller Design for Spacecraft Constrained Reorientation

In this subsection, the control law is designed to drive states of the spacecraft onto the sliding surface.

The following assumptions are made for the controller design:

Assumption a: Full states of the spacecraft are available.

Assumption b: External disturbances are unknown but bounded (i.e., $|d_i| < L_d$ with d_i being components of external disturbances $\bar{d}$) and the upper bound L_d is known.

Assumption c: The inertia matrix is constant and known.

Based on the potential field in Eq. (15) and the sliding surface in Eq. (14), an SMC law is proposed using the reaching law method [20] as

$$\bar{u} = \bar{\omega}^{\times} I\bar{\omega} - \alpha I \mathrm{vec}\left(\frac{d}{dt}(\boldsymbol{q}^* \otimes \nabla V_p)\right) - \sigma \mathrm{sgn}(\bar{s}) - k\bar{s} \tag{23}$$

where α is the parameter of the sliding surface in Eq. (14); both σ and k are parameters to be set with $\sigma > L_d$ and $k > 0$; $\mathrm{sgn}(\bar{s}) = [\mathrm{sgn}(s_1), \mathrm{sgn}(s_2), \mathrm{sgn}(s_3)]^T$ with $s_i (i = 1, 2, 3)$ being components of $\bar{s}$; and the term $(d/dt)(\boldsymbol{q}^* \otimes \nabla V_p)$ is calculated as (see Appendix E for the derivation)

$$\frac{d}{dt}(\boldsymbol{q}^* \otimes \nabla V_p) = -\frac{1}{2}\boldsymbol{\omega} \otimes \boldsymbol{q}^* \otimes \nabla V_p + \frac{1}{2}\boldsymbol{q}^* \otimes [(\nabla^2 V_p)(\boldsymbol{q} \otimes \boldsymbol{\omega})] \tag{24}$$

Theorem: Consider the model of rigid body rotation in Eqs. (1) and (5). Assume that, for the potential field V_p in Eq. (15), parameters are set so that there is no solution to $\nabla V_p = \mathbf{0}$ for all feasible $\boldsymbol{q}$ and the ratio β_1/β_2 is set as a suitably large positive value. Under Assumptions a, b, and c, the controller in Eq. (23) based on the sliding surface in Eq. (14) ensures that the states of the spacecraft asymptotically converge to $\{\boldsymbol{q} = \pm\boldsymbol{q}_d, \bar{\omega} = \bar{0}\}$.

Proof: First, we prove that the surface $\bar{s} = \bar{0}$ can be reached in finite time.

Consider a candidate Lyapunov function $V = (1/2)\bar{s}^T I\bar{s}$. V is a valid candidate Lyapunov function since $V \geq 0$ for all $\bar{s}$ and $V = 0$ if and only if $\bar{s} = \bar{0}$. There exist two positive constants $\lambda_{\min}$ and $\lambda_{\max}$ so that $(1/2)\lambda_{\min}\bar{s}^T\bar{s} \leq V \leq (1/2)\lambda_{\max}\bar{s}^T\bar{s}$. Thus, $\bar{s}^T\bar{s} \geq (2/\lambda_{\max})V$.

Taking the time derivative of V, and using the dynamic equation in Eq. (5) with the control law in Eq. (23), it yields

$$\dot{V} = \bar{s}^T\left(\bar{d} - \sigma \mathrm{sgn}(\bar{s}) - k\bar{s}\right) \tag{25}$$

Note that $\bar{s}^T(\bar{d} - \sigma \mathrm{sgn}(\bar{s})) \leq -a\bar{s}^T \mathrm{sgn}(\bar{s})$ with $a = \min\{\sigma - d_i\} > 0$. In addition, it holds that

$$\bar{s}^T \mathrm{sgn}(\bar{s}) \geq \sqrt{\bar{s}^T\bar{s}} \geq \sqrt{\frac{2}{\lambda_{\max}}}V^{1/2}$$

Thus, Eq. (25) yields

$$\dot{V} \leq -k\bar{s}^T\bar{s} - a\bar{s}^T \mathrm{sgn}(\bar{s}) \quad \leq -\frac{2k}{\lambda_{\max}}V - \frac{\sqrt{2}a}{\sqrt{\lambda_{\max}}}V^{1/2} \tag{26}$$

It follows that

$$\dot{V} + \frac{2k}{\lambda_{\max}}V + \frac{\sqrt{2}a}{\sqrt{\lambda_{\max}}}V^{1/2} \leq 0$$

According to the lemma for the finite-time stability introduced in Appendix F, the Lyapunov function V can reach 0 in finite time. Thus, the sliding variable $\bar{s}$ can reach $\bar{0}$ in finite time.

Next, we prove that the control objective can be achieved in the sliding mode.

When the sliding mode is reached, i.e., $\bar{s} = \bar{\omega} + \alpha \mathrm{vec}(\boldsymbol{q}^* \otimes \nabla V_p) = \bar{0}$, $\bar{\omega} = -\alpha \mathrm{vec}(\boldsymbol{q}^* \otimes \nabla V_p)$ holds. Based on Eq. (10), it yields $\dot{V}_p = -(1/2\alpha)\bar{\omega}^T\bar{\omega} \leq 0$. As there is no solution to $\nabla V_p = \mathbf{0}$ and the ratio β_1/β_2 is set as a suitably large positive value, according to the analysis of critical points in Sec. IV.B, the states will asymptotically converge to $\{\boldsymbol{q} = \pm\boldsymbol{q}_d, \bar{\omega} = \bar{0}\}$. (Note that $\boldsymbol{q}_d$ and $-\boldsymbol{q}_d$ are the same attitude.) This completes the proof. □

To avoid the chattering problem resulting from the use of the sign function, the sign function is replaced by the saturation function [28,29]. The ith component of $\bar{s}$ is replaced by $\mathrm{sat}(s_i, \epsilon)$ ($i = 1, 2, 3$). The saturation function is defined as [19]

$$\mathrm{sat}(s_i, \epsilon) = \begin{cases} 1, & s_i > \epsilon \\ \dfrac{s_i}{\epsilon}, & s_i \leq \epsilon \\ -1, & s_i < -\epsilon \end{cases} \tag{27}$$

where ϵ is a small positive constant. With the saturation function, the sliding variable can only converge to a neighborhood of the sliding surface [29].

Remark 1: The Lyapunov function V is bounded based on the analysis in the proof [as shown in Eq. (26)]. Thus, the sliding variable $\bar{s}$ is bounded. According to Eq. (14), the boundedness of $\bar{s}$ implies that ∇V_p is bounded. Based on the expression of ∇V_p in Eq. (18), it can be concluded that both f_F and f_M are bounded, which means that during the maneuver both attitude-constrained zones are not violated.

Remark 2: As long as the attitude does not start from a local minimum of the potential field, the attitude of the spacecraft will

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Table 2 Settings of attitude-constrained zones

Constrained zone	Constrained direction	Angle, deg
F1	$[0.2530, -0.3700, -0.8939]^T$	25
F2	$[0, 0.7071, 0.7071]^T$	40
F3	$[-0.8532, 0.4361, -0.2861]^T$	30
F4	$[0.9577, -0.2562, 0.1311]^T$	20
M	$[0.5871, 0.7520, -0.2996]^T$	60

Table 3 Controller parameters

Controller	Parameters
Proposed controller	$k_1 = 1, k_2 = 1, \beta_1 = 5\beta_2, \beta_2 = 0.01$
	$\alpha = 2, \sigma = 0.35, k = 4, \epsilon = 0.0001$
APF-based PD	$\alpha = 20, k_1 = 1.4, k_2 = 1.4$

not be trapped at an undesired attitude since the Lyapunov function decreases monotonically. Thus, as long as it holds that $\bar{s} \neq \bar{0}$ at the beginning of the maneuver, the problem of trapping at a local minimum can be avoided in the reaching phase. (In the SMC, the reaching phase is the phase before the sliding mode phase.)

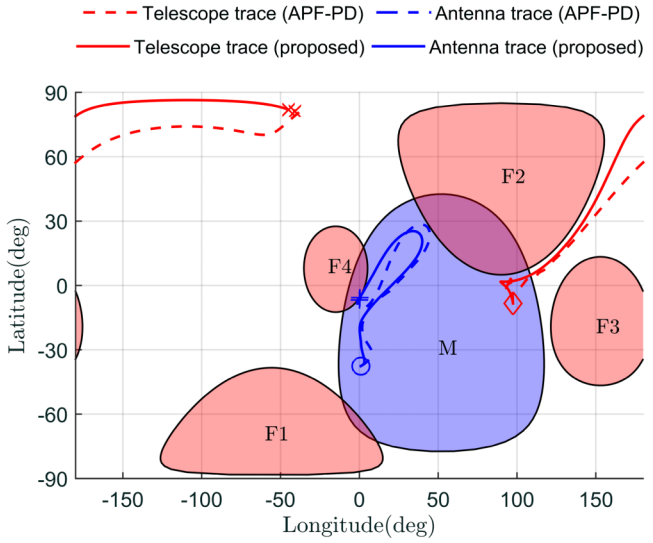


Fig. 3 Two-dimensional projection of traces of boresight vectors of telescope and antenna.

Remark 3: Compared with the sliding surface in [15], the proposed sliding surface in Eq. (14) has a novel property that, once the sliding mode is reached, the attitude of the spacecraft will converge to the desired one while satisfying attitude constraints automatically even under external disturbances. As attitude constraints are considered in the sliding surface design, the standard SMC method can be implemented without any adjustment.

Remark 4: The proposed controller in Eq. (23) needs the information of the true inertia matrix of the spacecraft, which is difficult to obtain in practice. For implementation purposes, the true inertia matrix I in Eq. (23) can be replaced with the estimated inertia matrix $\hat{I}$. Let $\Delta I = \hat{I} - I$ denote the model uncertainty. Again, consider a candidate Lyapunov function $V = (1/2)\bar{s}^T I \bar{s}$. With model uncertainties, taking the time derivative of V, it yields

$$\dot{V} = \bar{s}^T\left(\bar{d}_{\text{lumped}} - \sigma \text{sgn}(\bar{s}) - k\bar{s}\right) \tag{28}$$

where

$$\bar{d}_{\text{lumped}} = \bar{d} + \bar{\omega}^{\times}\Delta I\bar{\omega} - \alpha\Delta I \text{vec}\left[\frac{d}{dt}(\boldsymbol{q}^* \otimes \nabla V_p)\right]$$

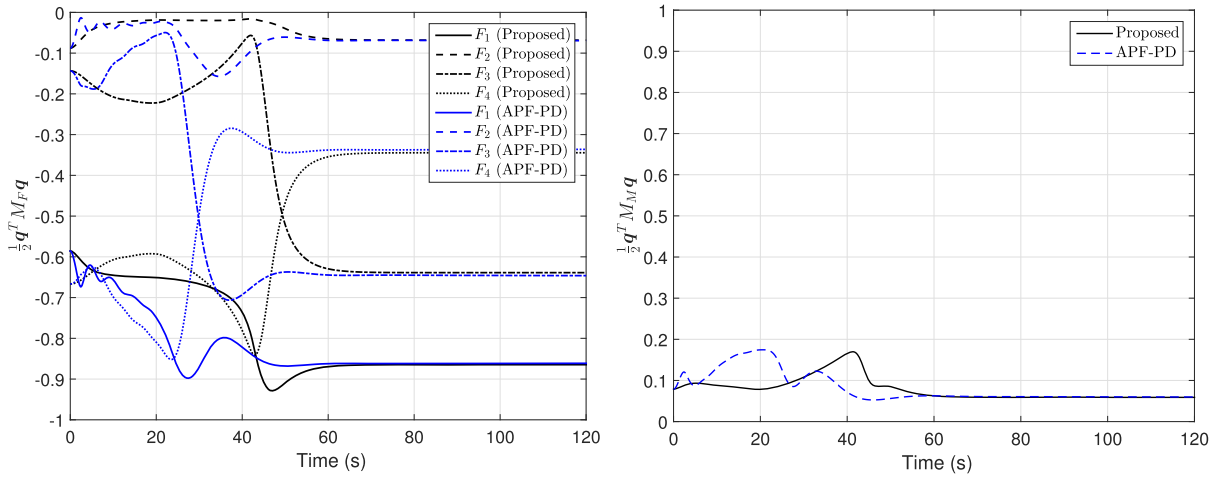
is the lumped disturbance that includes both external disturbances and the model uncertainty. When the terms introduced by the model uncertainty, i.e.,

$$\bar{\omega}^{\times}\Delta I\bar{\omega} - \alpha\Delta I \text{vec}\left[\frac{d}{dt}(\boldsymbol{q}^* \otimes \nabla V_p)\right]$$

are small, a properly large value of σ can deal with the lumped disturbance. However, when the terms introduced by the model uncertainty are large, the reaching phase or the sliding mode will be influenced. To consider the model uncertainty explicitly, adaptive control methods proposed in literature (e.g., [30]) can be applied.

V. Numerical Simulation

In this section, we demonstrate the effectiveness of the proposed APF-based SMC for spacecraft constrained reorientation and compare the proposed controller with the APF-based PD controller by Lee and Mesbahi [13]. The spacecraft in the simulation is equipped with a telescope and an antenna. The telescope and the antenna are aligned along the positive Z axis and the positive X axis, respectively, both in the body frame. The true inertia matrix of the spacecraft in the simulation is



a) The value of $\frac{1}{2}q^T M_F q$ **b) The value of $\frac{1}{2}q^T M_M q$**

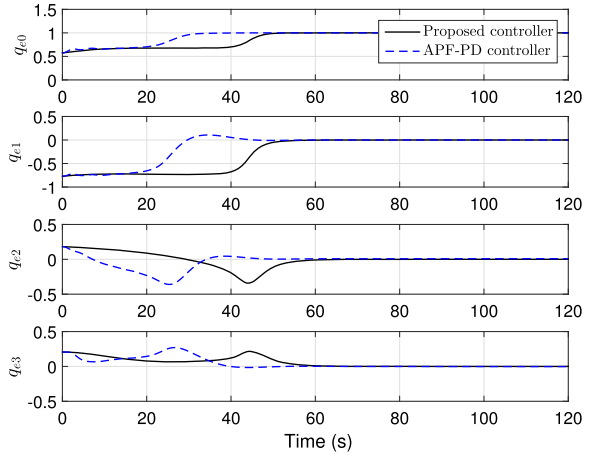
Fig. 4 Time histories of $(1/2)q^T M_F q$ and $(1/2)q^T M_M q$ under proposed controller and APF-based PD controller.

$$I = \begin{bmatrix} 60 & 5 & 1 \\ 5 & 50 & 2 \\ 1 & 2 & 70 \end{bmatrix}$$

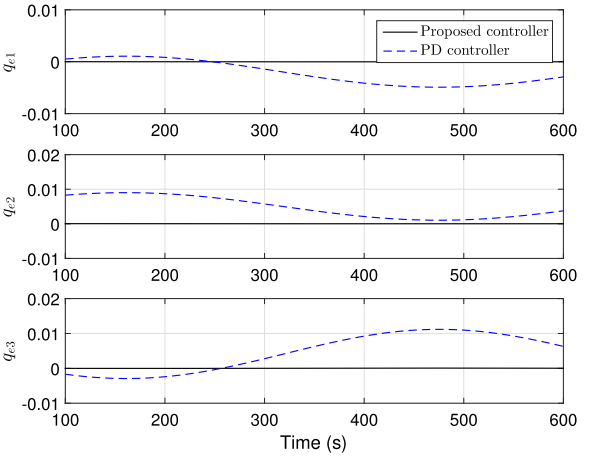
The estimated inertia matrix used in the proposed controller is set as $\hat{I} = \text{diag}(50, 45, 80)$ kg · m^2. The spacecraft is required to retarget its telescope from the initial orientation to the desired orientation. The initial and desired attitudes are set as $\boldsymbol{q}(0) = [0.6026, -0.73, 0.2, 0.253]^T$ and $\boldsymbol{q}_d = [0.9975, 0.0499, 0.0499, 0]^T$, respectively. As a rest-to-rest reorientation maneuver is considered, the initial and desired angular velocities are $\bar{\omega}(0) = \bar{\omega}_d = \bar{0}$. For the simulation, external disturbances are set as

$$\bar{d} = \begin{bmatrix} -0.02 + 0.03\sin(0.01t) \\ 0.05 + 0.04\sin(0.01t) \\ 0.04\text{–}0.07\sin(0.01t) \end{bmatrix} \text{N} \cdot \text{m}$$

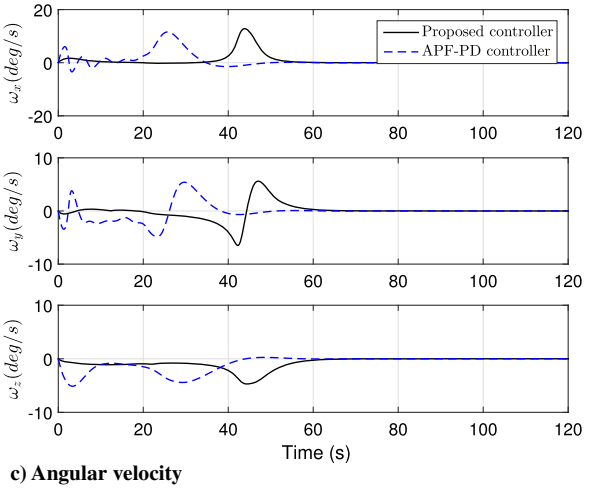
In the simulation, we consider a scenario with four attitude forbidden zones and one attitude mandatory zone. Settings of attitude-constrained zones are listed in Table 2. Parameters for the proposed controller and the APF-based PD controller are listed in Table 3.



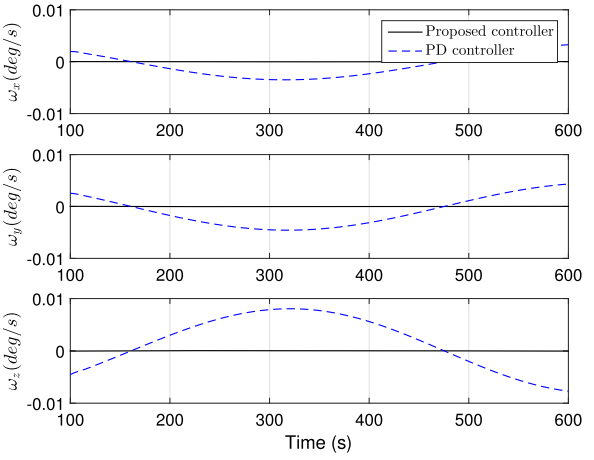
a) Error quaternion



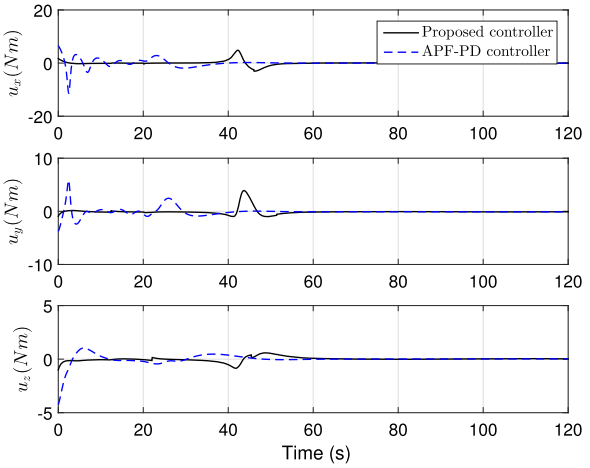
b) Vector part of error quaternion (zoom in)



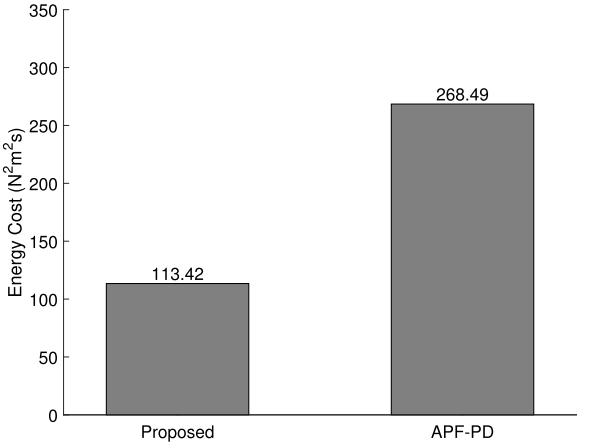
c) Angular velocity



d) Angular velocity (zoom in)



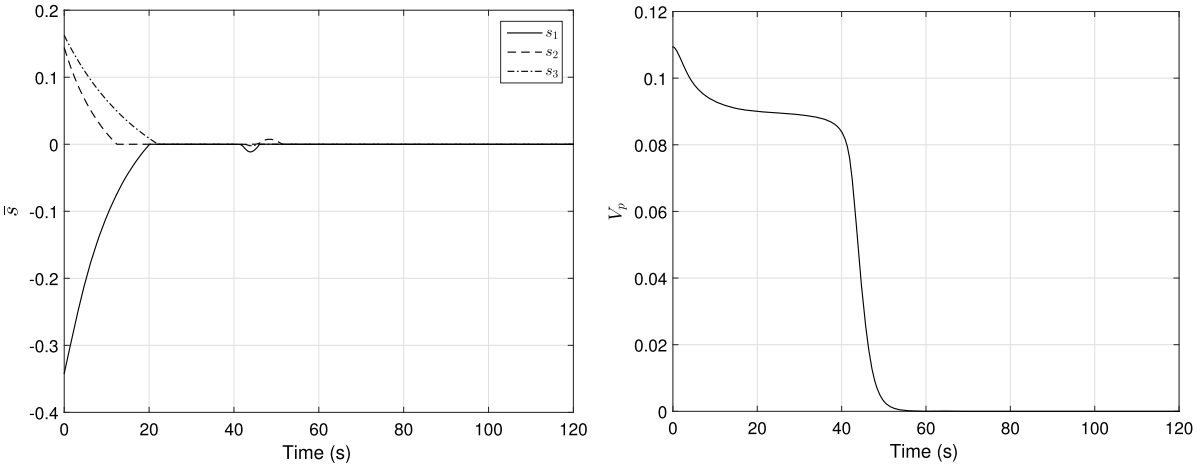
e) Control torque



f) Energy cost

Fig. 5 Time histories of error quaternion, angular velocity, control torque, and energy cost under proposed controller and APF-based PD controller.

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a) Sliding variable **b) The value of potential field**

Fig. 6 Time histories of sliding variable and value of potential field under proposed controller.

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Parameters for controllers are set so that maneuver settling times under both controllers are similar.

Figure 3 presents 2D projections of traces of boresight vectors of the telescope and the antenna on the unit celestial sphere under both controllers. Red- and blue-shaded areas indicate attitude forbidden zones and the attitude mandatory zone, respectively. Red diamond and cross denote the start and the end of the telescope boresight vector trace, respectively. Blue circle and plus denote the start and the end of the antenna boresight vector trace, respectively. Under both controllers, both telescope boresight vector traces stay outside all attitude forbidden zones and both antenna boresight vector traces stay within the attitude mandatory zone. Thus, attitude constraints are not violated during the reorientation. It is noted that endpoints of both telescope boresight vector traces are different. This is caused by external disturbances in the simulation. With external disturbances, under the APF-based PD controller, the telescope cannot be targeted toward the desired direction precisely. However, the proposed controller can retarget the telescope precisely even with external disturbances.

The time histories of the values of $(1/2)\boldsymbol{q}^T M_F \boldsymbol{q}$ and $(1/2)\boldsymbol{q}^T M_M \boldsymbol{q}$ are depicted in Fig. 4. The values of $(1/2)\boldsymbol{q}^T M_F \boldsymbol{q}$ for four attitude forbidden zones are always negative (Fig. 4a) and the values of $(1/2)\boldsymbol{q}^T M_M \boldsymbol{q}$ for the attitude mandatory zone are always positive (Fig. 4b). This verifies that all attitude constraints are satisfied during the rotation under both the APF-based PD and the proposed controller. Note that the values of $(1/2)\boldsymbol{q}^T M_F \boldsymbol{q}$ for attitude forbidden zone F2 under both controllers are very close to 0 during 0–25 s, which is consistent with the observation that some parts of both telescope boresight vector traces are very close to F2 in Fig. 3.

Figure 5 plots the time histories of the error quaternion, the angular velocity, the control torque, and the energy cost under the proposed and the APF-based PD controller. As shown in Fig. 5a, the error quaternion $\boldsymbol{q}_e$ converges to $[1, 0, 0, 0]^T$ under both controllers, which means that the telescope is retargeted to the desired orientation. The settling times under both controllers are similar, about 70 s. However, the control accuracy with external disturbances under the proposed controller is much better than that under the APF-based PD controller (as shown in Figs. 5b and 5d). As shown in Figs. 5c and 5e, under both the proposed and the APF-based PD controller, both the angular velocity and the control torque show peaks in their components. These peaks are caused by the fact that the telescope boresight vector approaches attitude forbidden zone F2. To avoid violating attitude constraints, repulsive torques are generated by the potential field to push away the telescope. Figure 5f plots the energy cost of a total simulation time of 150 s under the proposed and the APF-based PD controller. The energy cost is defined as

$$E(t) = \int_0^t \|\bar{u}\|^2 \, \mathrm{d}\tau$$

As shown in Fig. 5f, the energy cost under the proposed controller is 113.42 $\mathrm{N}^2 \cdot \mathrm{m}^2 \cdot \mathrm{s}$, which is less than half of that under the APF-based PD controller (268.49 $\mathrm{N}^2 \cdot \mathrm{m}^2 \cdot \mathrm{s}$). The big difference in energy cost can be explained as follows. Under the APF-based PD controller, at the beginning, the spacecraft has a faster response (see large values of components of the angular velocity at the beginning in Fig. 5c) than that under the proposed controller as larger torques are commanded by the APF-based PD controller (see Fig. 5e). Such a fast response leads to a large control effort for the APF-based PD controller to stop the telescope boresight vector from approaching attitude forbidden zone F2. In contrast, the initial response under the proposed controller is slow as small torques are commanded at the beginning. Thus, small control efforts are required to avoid the telescope boresight vector entering attitude forbidden zone F2.

Figure 6 plots the time histories of the sliding variable and the value of the potential field. As shown in Fig. 6a, components of the sliding variable can converge to 0 in finite time. The small oscillations of components of the sliding variable during 40–50 s result from the model uncertainty as discussed in Remark 4. The value of the potential field decreases monotonically (see Fig. 6b), where the sharp drop of the value during about 40–50 s corresponds to peaks of components of the angular velocity.

VI. Conclusions

This Note presents a potential field-based sliding surface design principle for spacecraft attitude maneuvers. Based on the proposed design principle, new results of sliding surfaces using different attitude parameterizations are obtained, which are shown to be general forms of existing classical sliding surfaces for attitude maneuvers in the literature. Furthermore, the proposed design principle is applied to the sliding surface design for spacecraft reorientation with attitude constraints (both attitude forbidden and mandatory zones). The novelty of the proposed sliding surface is that, once the sliding mode is reached, the attitude of the spacecraft will converge to the desired one while satisfying attitude constraints automatically. The existence and the characteristics of critical points of the potential field in the sliding mode are analyzed. As attitude constraints are considered in the sliding surface design, a standard controller design method of SMC is implemented for the constrained reorientation without any adjustment. The stability of the closed-loop system under the proposed APF-based sliding mode controller is proven using the Lyapunov method. Numerical simulations are presented to demonstrate the effectiveness and the performance of the proposed APF-based SMC for spacecraft constrained reorientation.

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Appendix A: Sliding Surface Design Based on Rodrigues Parameters

Define the potential field V_p as a function of $\bar{\rho}_e = [\rho_{e1}, \rho_{e2}, \rho_{e3}]^T$ with the definition $\bar{\rho}_e = \bar{\rho} - \bar{\rho}_d$. V_p is designed to have the property that $V_p \geq 0$ and $V_p = 0$ if and only if $\bar{\rho}_e = \bar{0}$. Taking the time derivative of V_p and using the kinematics in Eq. (2) yields

$$\begin{aligned}\dot{V}_p &= (\nabla V_p)^T \dot{\bar{\rho}}_e = (\nabla V_p)^T \left(\dot{\bar{\rho}} - \dot{\bar{\rho}}_d\right) \\ &= (\nabla V_p)^T \left[T(\bar{\rho})\bar{\omega} - T(\bar{\rho}_d)\bar{\omega}_d\right] \end{aligned} \tag{A1}$$

where

$$\nabla V_p \triangleq \frac{\partial V_p}{\partial \bar{\rho}_e} = \left[\frac{\partial V_p}{\partial \rho_{e1}}, \frac{\partial V_p}{\partial \rho_{e2}}, \frac{\partial V_p}{\partial \rho_{e3}}\right]^T$$

The following two kinds of analysis can be used for the sliding surface design.

Analysis 1: If we let $T(\bar{\rho})\bar{\omega} - T(\bar{\rho}_d)\bar{\omega}_d = -K_\alpha \nabla V_p$ (i.e., $\bar{\omega} = T^{-1}(\bar{\rho})T(\bar{\rho}_d)\bar{\omega}_d - T^{-1}(\bar{\rho})K_\alpha \nabla V_p$), with $K_\alpha \in \mathbb{R}^{3\times 3}$ being a diagonal positive matrix, it yields that $\dot{V}_p = -(\nabla V_p)^T K_\alpha \nabla V_p \leq 0$. When V_p decreases to zero, the desired attitude is reached. Define $\bar{\omega}_e \triangleq \bar{\omega} - T^{-1}(\bar{\rho})T(\bar{\rho}_d)\bar{\omega}_d$. The sliding surface is designed as

$$\bar{s}_1 = \bar{\omega}_e + T^{-1}(\bar{\rho})K_\alpha \nabla V_p \tag{A2}$$

where $K_\alpha \in \mathbb{R}^{3\times 3}$ is a diagonal positive matrix.

When the potential function is chosen as $V_p = (1/2)\|\bar{\rho}_e\|^2 = (1/2)\bar{\rho}_e^T \bar{\rho}_e$, the sliding surface in Eq. (A2) is simplified as $\bar{s}_1 = \bar{\omega}_e + T^{-1}(\bar{\rho})K_\alpha \bar{\rho}_e$, which is the same as the sliding surface in [24].

Analysis 2: With the definition of $\bar{\omega}_e$, the time derivative of V_p in Eq. (A1) can be rewritten as

$$\dot{V}_p = (\nabla V_p)^T T(\bar{\rho})\bar{\omega}_e \tag{A3}$$

Note that $T(\bar{\rho}) \geq (1/2)I_{3\times 3}$ [25]. If we let $\bar{\omega}_e = -c\nabla V_p$ with $c \in \mathbb{R}$ and $c > 0$, it yields that $\dot{V}_p = -c(\nabla V_p)^T T(\bar{\rho})\nabla V_p \leq -(1/2)c(\nabla V_p)^T \nabla V_p \leq 0$. When V_p decreases to zero, the desired attitude is reached. Based on this observation, the sliding surface is designed as

$$\bar{s}_2 = \bar{\omega}_e + c\nabla V_p \tag{A4}$$

where c is a positive scalar.

When the potential function is chosen as $V_p = (1/2)\bar{\rho}_e^T \Lambda \bar{\rho}_e$, where $\Lambda \in \mathbb{R}^{3\times 3}$ is a diagonal positive matrix, the sliding surface in Eq. (A4) is simplified as $\bar{s}_2 = \bar{\omega}_e + \Lambda' \bar{\rho}_e$, with $\Lambda' = c\Lambda$, which is the same as the sliding surface in [25].

Appendix B: Sliding Surface Design Based on Modified Rodrigues Parameters

Define the potential field V_p as a function of $\bar{\sigma}_e = [\sigma_{e1}, \sigma_{e2}, \sigma_{e3}]^T$ with the definition $\bar{\sigma}_e = \bar{\sigma} - \bar{\sigma}_d$. V_p is designed to have the property that $V_p \geq 0$ and $V_p = 0$ if and only if $\bar{\sigma}_e = \bar{0}$. Taking the time derivative of V_p and using the kinematics in Eq. (4) yields

$$\begin{aligned}\dot{V}_p &= (\nabla V_p)^T \dot{\bar{\sigma}}_e = (\nabla V_p)^T (\dot{\bar{\sigma}} - \dot{\bar{\sigma}}_d) \\ &= (\nabla V_p)^T [F(\bar{\sigma})\bar{\omega} - F(\bar{\sigma}_d)\bar{\omega}_d] \end{aligned} \tag{B1}$$

where

$$\nabla V_p \triangleq \frac{\partial V_p}{\partial \bar{\sigma}_e} = \left[\frac{\partial V_p}{\partial \sigma_{e1}}, \frac{\partial V_p}{\partial \sigma_{e2}}, \frac{\partial V_p}{\partial \sigma_{e3}}\right]^T$$

If we let $F(\bar{\sigma})\bar{\omega} - F(\bar{\sigma}_d)\bar{\omega}_d = -K_\alpha \nabla V_p$ (i.e., $\bar{\omega} = F^{-1}(\bar{\sigma})F(\bar{\sigma}_d)\bar{\omega}_d - F^{-1}(\bar{\sigma})K_\alpha \nabla V_p$), with $K_\alpha \in \mathbb{R}^{3\times 3}$ being a diagonal positive matrix, it yields that $\dot{V}_p = -(\nabla V_p)^T K_\alpha \nabla V_p \leq 0$. When V_p decreases to zero, the desired attitude is reached. Define $\bar{\omega}_e \triangleq \bar{\omega} - F^{-1}(\bar{\sigma})F(\bar{\sigma}_d)\bar{\omega}_d$. The sliding surface can be designed as

$$\bar{s} = \bar{\omega}_e + F^{-1}(\bar{\sigma})K_\alpha \nabla V_p \tag{B2}$$

where $K_\alpha \in \mathbb{R}^{3\times 3}$ is a diagonal positive matrix.

The potential function can be chosen as $V_p = (1/2)\|\bar{\sigma}_e\|^2 = (1/2)\bar{\sigma}_e^T \bar{\sigma}_e$. Then, one has $\nabla V_p = \bar{\sigma}_e$. Thus, the sliding surface in Eq. (B2) is simplified as $\bar{s} = \bar{\omega}_e + F^{-1}(\bar{\sigma})K_\alpha \bar{\sigma}_e$. Note that the sliding surface in [26] is a special result of the general form in this Note.

Appendix C: Related Calculation for Potential Field in Eq. (15)

The first partial derivatives $\partial f_F/\partial \boldsymbol{q}$ and $\partial f_M/\partial \boldsymbol{q}$ are calculated by

$$\frac{\partial f_F}{\partial \boldsymbol{q}} = \sum_{j=1}^{m}\sum_{i=1}^{n} -\frac{2k_1}{\boldsymbol{q}^T M_{Fi}^j \boldsymbol{q}} (\boldsymbol{q}^T M_{Fi}^j)^T \tag{C1}$$

and

$$\frac{\partial f_M}{\partial \boldsymbol{q}} = -\frac{2k_2}{\boldsymbol{q}^T M_M \boldsymbol{q}} (\boldsymbol{q}^T M_M)^T \tag{C2}$$

The second partial derivatives $\partial^2 f_F/\partial \boldsymbol{q}^2$ and $\partial^2 f_M/\partial \boldsymbol{q}^2$ are calculated by

$$\frac{\partial^2 f_F}{\partial \boldsymbol{q}^2} = \sum_{j=1}^{m}\sum_{i=1}^{n}\left[\frac{4k_1}{(\boldsymbol{q}^T M_{Fi}^j \boldsymbol{q})^2} (\boldsymbol{q}^T M_{Fi}^j)^T \boldsymbol{q}^T M_{Fi}^j - \frac{2k_1}{\boldsymbol{q}^T M_{Fi}^j \boldsymbol{q}} M_{Fi}^j\right] \tag{C3}$$

and

$$\frac{\partial^2 f_M}{\partial \boldsymbol{q}^2} = \frac{4k_2}{(\boldsymbol{q}^T M_M \boldsymbol{q})^2} (\boldsymbol{q}^T M_M)^T \boldsymbol{q}^T M_M - \frac{2k_2}{\boldsymbol{q}^T M_M \boldsymbol{q}} M_M \tag{C4}$$

The Hessian matrix $\nabla^2 V_p = [\partial(\nabla V_P)/\partial \boldsymbol{q}]$ is as follows:

$$\begin{aligned}\nabla^2 V_p = &-2\beta_2 \delta(q_{e0})\left[\frac{\partial f}{\partial \boldsymbol{q}} \boldsymbol{q}_d^T + \boldsymbol{q}_d \left(\frac{\partial f}{\partial \boldsymbol{q}}\right)^T\right] \\ &+ 2\beta_2 (1 - \delta(q_{e0})\boldsymbol{q}_d^T \boldsymbol{q}) \frac{\partial^2 f}{\partial \boldsymbol{q}^2} \end{aligned} \tag{C5}$$

where

$$\frac{\partial^2 f}{\partial \boldsymbol{q}^2} = \frac{\partial^2 f_F}{\partial \boldsymbol{q}^2} + \frac{\partial^2 f_M}{\partial \boldsymbol{q}^2}$$

Appendix D: Proof of Proposition 1

First, we prove that $\boldsymbol{q}^* \otimes \nabla V_p = \boldsymbol{0}$ yields $\nabla V_p = \boldsymbol{0}$.

If $\nabla V_p \neq \boldsymbol{0}$, then the identity $\boldsymbol{q}^* \otimes \nabla V_p = \boldsymbol{0}$ can be rewritten as $\|\nabla V_p\|[\boldsymbol{q}^* \otimes (\nabla V_p/\|\nabla V_p\|)] = \boldsymbol{0}$. As $\nabla V_p/\|\nabla V_p\|$ is a unit quaternion, $\boldsymbol{q}^* \otimes (\nabla V_p/\|\nabla V_p\|)$ is also a unit quaternion, i.e., $\|\boldsymbol{q}^* \otimes (\nabla V_p/\|\nabla V_p\|)\| = 1$. Thus, $\|\nabla V_p\|[\boldsymbol{q}^* \otimes (\nabla V_p/\|\nabla V_p\|)] = \boldsymbol{0}$ yields $\|\nabla V_p\| = 0$, which is contradictory to the assumption that $\nabla V_p \neq \boldsymbol{0}$. Thus, $\boldsymbol{q}^* \otimes \nabla V_p = \boldsymbol{0}$ yields $\nabla V_p = \boldsymbol{0}$.

Next, we prove that, based on Eq. (18), $\nabla V_p = \mathbf{0}$ yields $1 - \delta(q_{e0})\boldsymbol{q}_d^T\boldsymbol{q} \neq 0$, and hence $\boldsymbol{q} \neq \delta(q_{e0})\boldsymbol{q}_d$.

In fact, if $1 - \delta(q_{e0})\boldsymbol{q}_d^T\boldsymbol{q} = 0$, Eq. (18) can be simplified as $\nabla V_p = -2\delta(q_{e0})(\beta_1 + \beta_2 f)\boldsymbol{q}_d$. Since $\delta(q_{e0}) = \pm 1, \beta_1 > 0, \beta_2 > 0, f > 0$, and $\boldsymbol{q}_d$ is a unit quaternion, one has $\|\nabla V_p\| = 2(\beta_1 + \beta_2 f) > 0$, which is contradictory to the conclusion $\nabla V_p = \mathbf{0}$. Thus, based on Eq. (18), $\nabla V_p = \mathbf{0}$ yields $1 - \delta(q_{e0})\boldsymbol{q}_d^T\boldsymbol{q} \neq 0$. It is obvious that $1 - \delta(q_{e0})\boldsymbol{q}_d^T\boldsymbol{q} \neq 0$ yields $\boldsymbol{q} \neq \delta(q_{e0})\boldsymbol{q}_d$. This completes the proof of Proposition 1.

Appendix E: Derivation of Eq. (24)

$$\begin{aligned}\frac{d}{dt}(\boldsymbol{q}^* \otimes \nabla V_p) &= \dot{\boldsymbol{q}}^* \otimes \nabla V_p + \boldsymbol{q}^* \otimes \frac{d}{dt}(\nabla V_p) \\ &= -\frac{1}{2}\boldsymbol{\omega} \otimes \boldsymbol{q}^* \otimes \nabla V_p + \boldsymbol{q}^* \otimes (\nabla^2 V_p \dot{\boldsymbol{q}}) \\ &= -\frac{1}{2}\omega \otimes \boldsymbol{q}^* \otimes \nabla V_p + \frac{1}{2}\boldsymbol{q}^* \otimes [(\nabla^2 V_p)(\boldsymbol{q} \otimes \boldsymbol{\omega})]\end{aligned} \tag{E1}$$

where $\dot{\boldsymbol{q}}^* = -(1/2)\boldsymbol{\omega} \otimes \boldsymbol{q}^*$ (which can be obtained by differentiating $\boldsymbol{q}^* \otimes \boldsymbol{q} = \mathbf{1}$) and

$$\frac{d}{dt}(\nabla V_p) = \frac{\partial(\nabla V_P)}{\partial \boldsymbol{q}}\frac{d\boldsymbol{q}}{dt} = \nabla^2 V_p \dot{\boldsymbol{q}}$$

are used.

Appendix F: Lemma for Finite-Time Stability

Lemma [31]: A Lyapunov condition of the finite-time stability can be given as

$$\dot{V}(x) + \lambda_1 V(x) + \lambda_2 V^r(x) \leq 0, \quad (\lambda_1 > 0, \lambda_2 > 0, 0 < r < 1) \tag{F1}$$

and the settling time can be estimated by

$$T_s \leq \frac{1}{\lambda_1(1-r)}\ell_n\left[\frac{\lambda_1 V^{1-r}(x_0) + \lambda_2}{\lambda_2}\right] \tag{F2}$$

where $V(x_0)$ is the initial value of the Lyapunov function V.

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